Extracting Recipes from Cooking Videos using V-JEPA2 Embeddings

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Introduction

V-JEPA 2 presents itself as a innovative and accessible way to identify complex physical interactions in videos.
Understanding **egocentric cooking videos** is challenging: first-person footage captures natural kitchen activities but suffers from clutter, and difficult recognition of objects.

We propose a **two-stage pipeline**:

1. **Relevance filtering** — separate cooking actions from background events.
2. **Action classification** — fine-grained verb/noun recognition on relevant clips.

We exploit **V-JEPA2 embeddings** and train only lightweight MLP heads, keeping compute requirements minimal.

Dataset: EPIC-KITCHENS-100

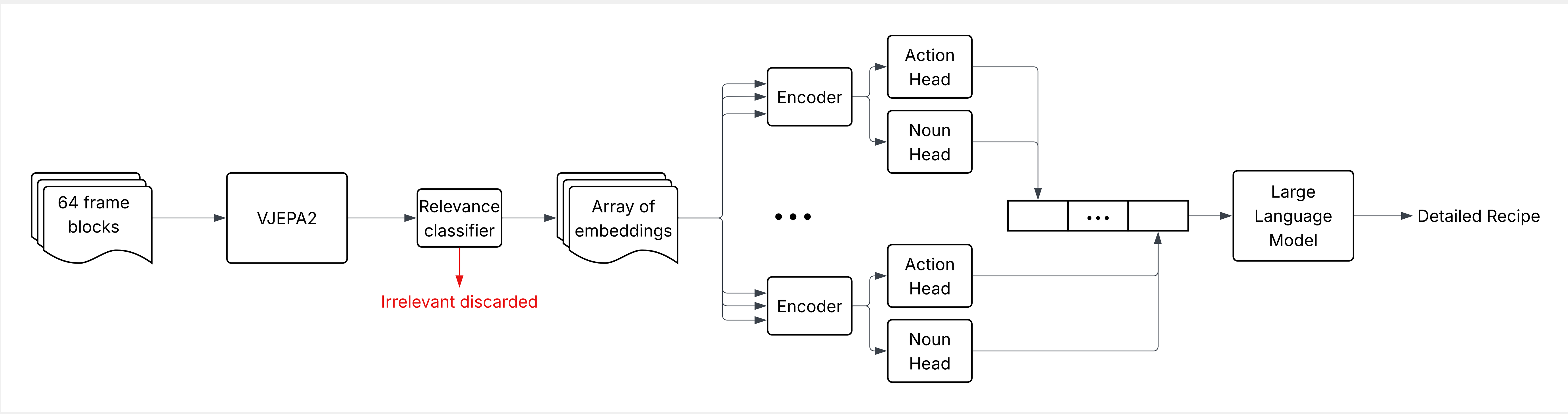
- **276 videos** (249 train / 27 test)
- First-person kitchen recordings
- Actions labeled with **126 verb** and **353 noun** classes
- Derivate binary relevance from actions and nouns

Split	Fraction
Relevant	≈65.7%
Irrelevant	≈34.3%

Why V-JEPA2-small?

We use the **small** variant of V-JEPA2 (ViT-L backbone, ≈307M parameters). Its compact footprint means the full pipeline can run on a **MacBook Pro** or any consumer laptop with sufficient GPU RAM — though embedding extraction remains slow on local hardware. For this project we used **cloud compute** to embed the full dataset in reasonable time.

Pipeline Overview



Classification Heads

Relevance Classifier — TBD

Action Classifier — temporal Transformer over a 7-block window (radius 3). Embeddings are projected 1024 → 768 with learned positional encodings, then processed by a 2-layer, 8-head TransformerEncoder. The center token is fed to a single linear head predicting the **action** (a noun–verb pair). A 3-seed ensemble averages logits at inference.

Why a Transformer Encoder?

A single block provides only a narrow temporal context. A Transformer encoder attends across the full sequence of block embeddings, aggregating multi-window evidence before an MLP head produces the final classification.

Results

Stage	Accuracy
Relevance	XX%
Action (verb)	XX%
Noun	XX%

LLM Post-Processing

The action classifier outputs a **sequence of (verb, noun) pairs** aligned to video blocks. We pass this sequence to **Llama 3.2** which synthesizes the raw event list into a fluent, human-readable recipe: coherent steps, resolved repetitions, and natural prose.

This bridges the gap between frame-level predictions and interpretable cooking instructions.

Conclusion

- **Two-stage design** (relevance + classification) cuts background noise before fine-grained prediction.
- **Transformer encoder** provides multi-block temporal context with minimal added parameters.
- **LLM post-processing** turns sparse event sequences into readable recipes.
- Pipeline runs on consumer hardware with no backbone tuning.